Solution Architecture

Date	15 July 2026
Team ID	SWTID-2026-2658
Project Name	Credit card Approval and Prediction System
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.

Presentation / Client Layer

HTML5, CSS3, Bootstrap 5, JavaScript (Flask Templates)

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API Gateway / Load Balancer

Flask Application Server (No separate API Gateway or Load Balancer used)

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Auth Service	Core Logic Service	External API Integrations
Not Used (No user authentication required)	Python, Flask, Scikit-learn Random Forest Model, Pandas, NumPy	**Not Used** (Prediction is performed locally using the trained model

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Data / Storage Layer

Joblib (`credit_card_model.pkl`) for trained model storage; CSV datasets (`application_record.csv`, `credit_record.csv`) used during model training

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides a user-friendly interface for applicants to enter personal, financial, and employment details and displays the credit card approval prediction result.	HTML5, CSS3, Bootstrap 5, JavaScript, Flask Templates
API Gateway	Acts as the entry point for handling HTTP requests from the user interface and routes them to the Flask backend for processing.	Flask Web Server
Microservices	Performs data preprocessing, loads the trained Random Forest model, processes applicant information, and generates the approval prediction.	Python, Flask, Scikit-learn, Pandas, NumPy, Joblib
Database / Storage	Stores the trained machine learning model used for prediction. The datasets are used only during model training and are not accessed during prediction.	Joblib (`credit_card_model.pkl`), CSV files (`application_record.csv`, `credit_record.csv`)